New trading relationships can change the dominant vehicle.

Centre for Blockchain Technologies

The Making of Dominant Vehicle Currencies

Evidence from DeFi

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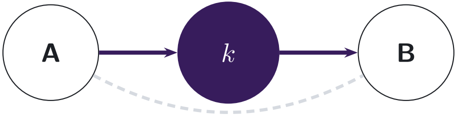
A cross-border payment needs someone to make the FX market

THE LIQUIDITY-PROVIDER PROBLEM

A payment provider receives one currency but must deliver another it does not hold.

Project Nexus requires an FX provider. Project Rialto uses a vehicle when direct exchange is unavailable.



Currency k bridges an illiquid pair.

vehicle currency

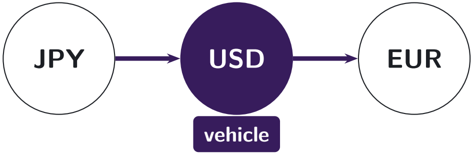
Note: Project Nexus states that every cross-border, cross-currency payment needs an actor willing and able to exchange the currencies. Project Rialto experimentally uses a vehicle currency when direct exchange is unavailable. The paper studies the intermediary asset observed inside DEX pool routes.

Pool routes reveal the vehicle currency

FX ANALOGY

Yen–euro may trade through dollars.

Conventional turnover data show two legs, not the customer's connected route.



Pool logs reveal the path.

DeFi is on-chain finance executed by smart contracts; decentralised exchanges (DEXs) trade tokens through liquidity pools. **Stablecoins** are tokens designed to track a reference currency, usually the US dollar.

Note: Traditional FX records show turnover in each currency pair, but not whether two legs belong to one customer route. The DEX setting preserves the transaction-level pool path, so the intermediary asset is observed directly.

The route panel links pool-level swaps

472 million
pool-level swaps

2,332
calendar dates

8
DEX deployments

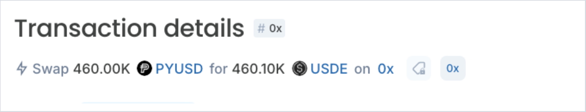
February 2020–June 2026

Uniswap v2, v3, and v4; SushiSwap v2 and v3; Curve; Balancer; and Fluid.

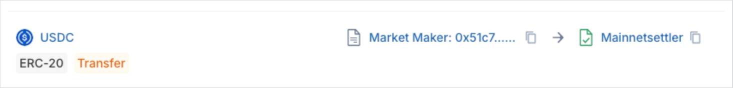
Note: The listed deployments form the principal Ethereum route panel. Uniswap v1 enters the separate protocol-architecture analysis because its retained records do not recover endpoint-pair identities.

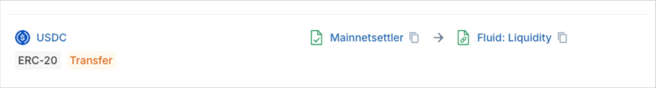
Pool data may start after the user instruction

USER INSTRUCTION RECORDED BY THE EXPLORER



THE DECISIVE SETTLEMENT TRANSFERS





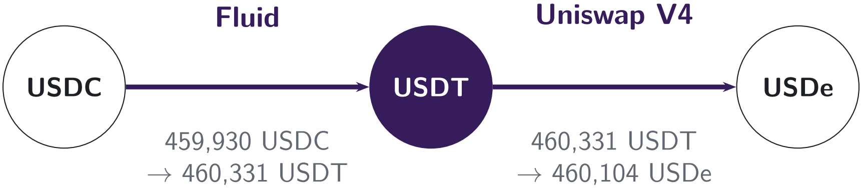
The maker supplies the USDC used by the Fluid leg. The instruction is PYUSD $\rightarrow$ USDe.

Note: Transaction `0xbda1d07f...9bf67ad9`, 10 January 2026. The data do not reveal whether PYUSD–USDC is another vehicle leg or executor inventory. Disconnected groups comprise 6.2% of reconstructed components in 2026. Treating each internally valid component as a separate route raises stablecoin count dominance from 42.3% to 43.7%.

Inside the pools, USDT links USDC to USDe

ORDERED TRANSFERS AND ROUTE RECONSTRUCTION

User instruction: PYUSD $\rightarrow$ USDe $\Longrightarrow$ Pool route: USDC $\rightarrow$ USDT $\rightarrow$ USDe



vehicle

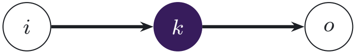
The observed endpoint pair is USDC $\rightarrow$ USDe. Its legs are USDC $\rightarrow$ USDT and USDT $\rightarrow$ USDe; USDT is the vehicle that links them.

Note: The route is the connected pool-swap component of transaction `0xbda1d07f...9bf67ad9` on 10 January 2026.

Vehicle dominance is an intermediary share

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k.
- Count dominance weights each route equally; value dominance weights comparable routed value.

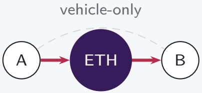


Count-weighted dominance
Share of indirect routes that pass through k

Value-weighted dominance
Share of comparable routed value that passes through k

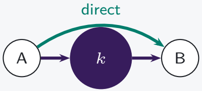
Architecture changes routing opportunities

V1 mandatory hub



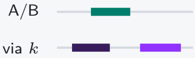
Hub use is imposed

V2 trading-pair choice



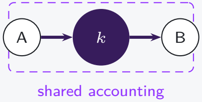
Hub use becomes a choice

V3 range choice



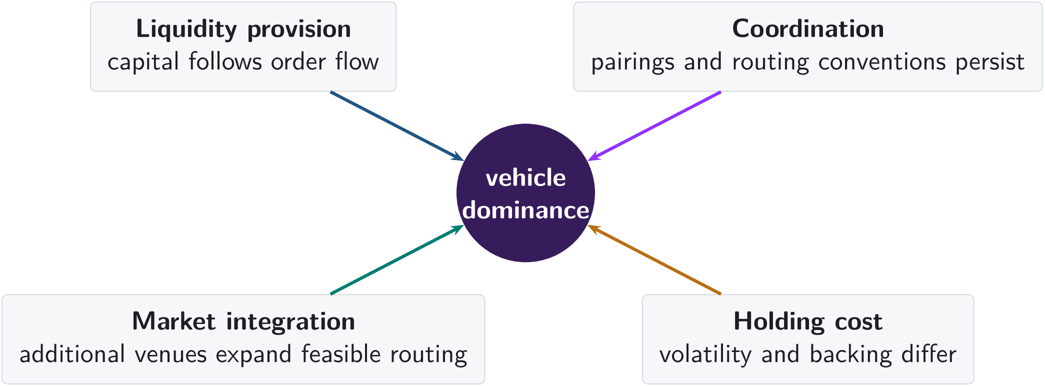
Trading opportunities shift

V4 net settlement



Settlement differs from route use

Four economic forces shape vehicle dominance



Native-asset pairing persists after it becomes optional

V1: PROTOCOL RULE

217,003 token-to-token trades pass through ETH

Every pool is an ETH–token pool, so the architecture imposes the vehicle.

$\Longrightarrow$

V2: MARKET CHOICE

95.5%

of single-leg V2 trades use a WETH pool in 2026

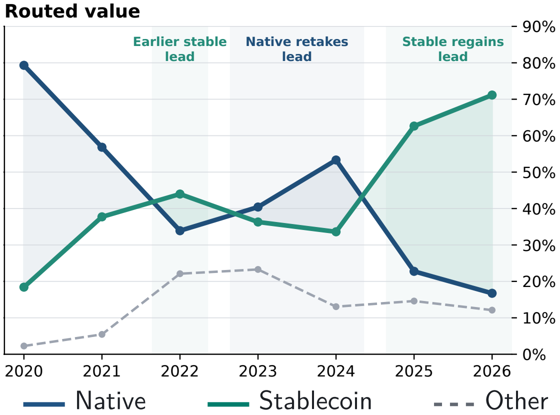
97.9%

of endpoint pairs first traded in 2026 include WETH

Note: V1 links token-to-ETH and ETH-to-token swaps within one transaction. V2 reports single-leg trades and newly traded endpoint pairs on that venue.

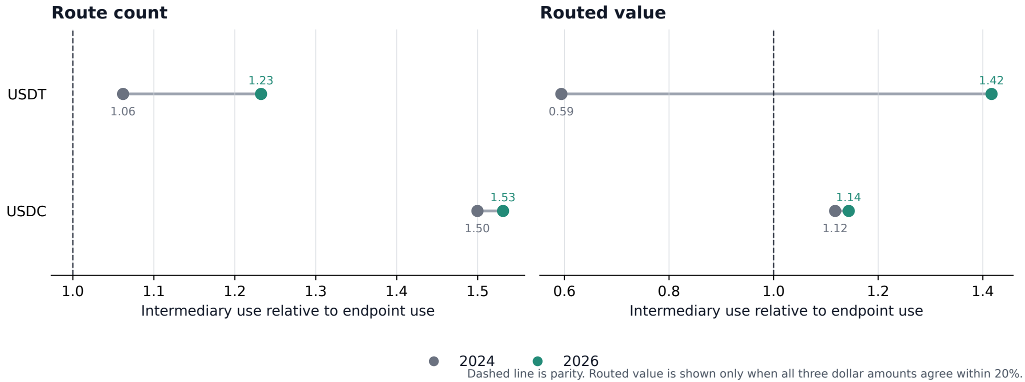
Stablecoins regain the routed-value lead by 2026

MATCHED JANUARY–JUNE PERIODS



Note: Exact two-leg routes. Routed value requires agreement among source, intermediary, and destination dollar values. The 2026 endpoint covers January through June.

USDT carries the stablecoin expansion



USDT crosses parity by routed value and extends its count advantage; USDC changes little.

Note: Excess use divides intermediary share by route-endpoint share on the same perimeter. The figure compares the same January through June dates in 2024 and 2026.

Endpoint demand predicts vehicle use within day

$$I_{a,t} = \alpha + \beta D_{a,t} + \mathbf{A}_a'\theta + \mu_t + \eta_a + \varepsilon_{a,t}$$

$I_{a,t}$ intermediary episode share

$D_{a,t}$ own endpoint-demand share

$\mathbf{A}_a$ asset-class indicators

μ_t date effects

η_a currency effects

Note: Outcome and demand are shares of the same day's route universe, measured in percentage points (pp). The estimates are descriptive within-day associations.

The demand relationship survives currency and date effects

	Intermediary share ($I_{a,t}$) [pp]			
	(1) Pooled	(2) Date FE	(3) + demand	(4) + currency FE
Native currency	$+34.56$	$+34.55$	$+0.05$	—
	(23.04)	(23.04)	(0.56)	—
Stablecoin	$+2.42^{*}$	$+2.42^{*}$	$+0.85^{*}$	—
	(1.32)	(1.32)	(0.50)	—
Endpoint-demand share ($D_{a,t}$)	—	—	$+1.59^{***}$	$+0.98^{***}$
	—	—	(0.03)	(0.30)
Date fixed effects	No	Yes	Yes	Yes
Currency fixed effects	No	No	No	Yes

Endpoint-demand share passes through almost one for one after both fixed effects.

Note: 1,828,862 currency-days and 2,259 dates. Parentheses report currency/date-clustered standard errors. Class coefficients are percentage-point differences; the demand coefficient is percentage points of intermediary share per percentage point of endpoint-demand share. Asterisks $*$, $**$, and $***$ denote significance at 10%, 5%, and 1%, respectively.

WETH-linked endpoint pairs carry the count rotation

WETH

WETH is an endpoint

↓

Eligible intermediaries: stablecoins

$+21.1$ **pp**

All matched endpoint pairs

WETH-linked route-count mass: 26.6% $\rightarrow$ 46.8%

$+3.7$ **pp**

Endpoint pairs without WETH endpoints

79,347 matched groups

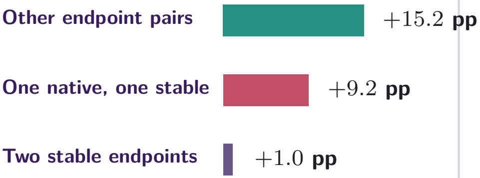
Changes within fixed endpoint pairs are $+0.2$ pp overall and $+0.3$ pp without WETH endpoints.

Note: Exact two-leg native-WETH-versus-stablecoin routes on common month-days. Within-endpoint-pair estimates fix endpoint pair, month-day, and route scope.

Who trades with whom makes the vehicle

CONTRIBUTION TO THE STABLECOIN-SHARE INCREASE

Route count: $+25.4$ pp



Routed value: $+43.9$ pp

USDT supplies $+13.7$ pp of the $+13.2$ pp two-stable endpoint value channel.

Note: 181 common days. Exact additive decomposition; descriptive.

Trading shifts toward stablecoin-heavy endpoint pairs

Without WETH endpoints

$+21.5$ **pp**

eligible stablecoin share of routed value

Full matched sample: $+27.2$ pp

$=$

Routed-value reallocation

$+23.5$ **pp**

Value shifts toward endpoint-pair groups with higher stablecoin shares.

$+$

Change within groups

-2.0 **pp**

Stablecoin share changes within the same endpoint-pair group.

Note: The daily midpoint identity is exact after removing WETH endpoints. Routed value requires agreement among source, intermediary, and destination dollar values.

New endpoint pairs deliver the largest positive component

EXACT POOLED DECOMPOSITION, 2024 TO 2026

2024 **16.9%** → 2026 **42.5%** $+25.7$ **pp**

aggregate rotation

Continuing endpoint pairs

-0.1 **pp** net within-endpoint-pair change

$+8.6$ **pp** activity reallocation

Endpoint-pair entry and exit

$+21.0$ **pp** newly traded endpoint pairs

-3.3 **pp** endpoint-pair exit

Pair entry contributes $+21.0$ pp; net switching inside continuing pairs contributes -0.1 pp.

Note: Matched January–June periods in 2024 and 2026. The within-pair component combines $+1.3$ pp toward stablecoins and -1.4 pp toward native assets. The remaining continuing-pair-weight component is -0.5 pp. Components sum using unrounded values.

Local bridge depth predicts route choice

84.1%

route share captured by the deepest bridge

2024 **75.5%**

2026 **86.5%**

deepest-bridge share

$+6.79$ **pp**

per log point of weaker-leg deposited capital

The weaker atomic leg carries the economically relevant depth constraint.

Note: The analysis compares five vehicle assets within locally supported endpoint-date-route-scope sets. Depth is $\log(1 +$ prior deposited capital in USD$)$ on the weaker atomic leg; the regression also controls for vehicle reach.

Stablecoin penetration rises with relative bridge depth

FIRST 30 DAYS AFTER STABLE-BRIDGE AVAILABILITY

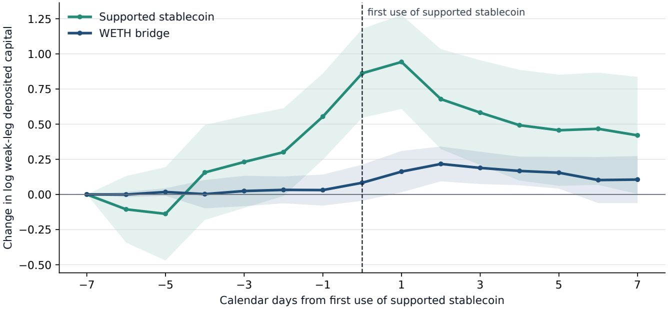
Below $0.1\times$ WETH depth	At least WETH depth	At least $2\times$ WETH depth
2.4%	**53.0%**	**69.9%**

First-month use of the supported stablecoin: 42.6% below $0.1\times$ WETH depth, 84.1% at or above it.

Note: The route-share cards use 11,327 endpoint-pair days across 855 events. Adoption gives equal weight to 853 events with positive stable and WETH depth and matches support, depth, and first use to the same stablecoin set. The adjusted gap is $+25.2$ pp (SE 5.8 pp); with two non-stablecoin endpoints, it is $+35.8$ pp (SE 11.9 pp). Capital and demand are jointly determined.

Bridge capital builds before use

SUPPORTED STABLECOIN FIRST USED AT DAY ZERO



$+0.86$

stablecoin bridge
day -7 to 0

$+0.78$

pre-route change
relative to WETH

-0.44

stablecoin bridge
day 0 to +7

Route-specific capital accumulates before use, then partially recedes.

Note: Equal-event path for 246 events whose supported stablecoin is first used 7 to 119 days after persistent support. Capital is prior-calendar V2-family deposited capital on the weaker leg; shading gives 95% confidence intervals clustered by ordered endpoint pair and adoption date. The sample conditions on eventual use and does not identify provider intent.

Most first-use capital sits in pools already active

SUPPORTED STABLECOIN, DAY -7 TO FIRST USE; MATCHED WETH BENCHMARK

92.5%

of first-use-day capital lies in pools active one week earlier

$+0.16$

continuing-pool capital growth relative to WETH [log points]

$+6.9$ **pp**

higher chance of a newly active pool relative to WETH

Continuing pools hold most capital; pool activation also rises before adoption.

Note: Equal-event estimates for 246 first-use events. A newly active pool has positive prior-calendar deposited capital on day zero and none on day -7; continuing pools are positive on both dates. Difference-column inference is two-way clustered by ordered endpoint pair and adoption date, with Holm adjustment across four pool-composition outcomes. Deposited-capital stocks do not identify provider cash flow or intent.

V4 internal routing forecasts later entry in volatile markets

10 PP MORE INTERNAL SAME-ASSET ROUTING

DAYS 1 TO 30

$+0.086^{***}$

actions by 180-day incumbent origins [log points]

s.e. 0.024

DAYS 31 TO 120

$+0.153^{***}$

origins first active after measurement [log points]

s.e. 0.032

PERSISTENT VOLATILITY

$+0.318^{***}$

additional first-active-origin response per 1 SD of 30-day WETH volatility [log points]

s.e. 0.039

Incumbents act within a month; first-active origins follow, especially when volatility stays high.

Note: 1,107 V4 vehicle-days and 223 date clusters; zero-liquidity updates excluded; 180 prior days required. Volatility is the lagged 30-day mean of realised WETH volatility. Vehicle and date effects; origin-day controls; vehicle and control volatility slopes; date-clustered standard errors. The interaction uses Holm adjustment across two outcomes. A same-vehicle-date V3 comparison does not distinguish the mature-calendar level or volatility slope. Origins proxy for account participation, not LP-position ownership. Predictive associations.

Dominance forms through entry, depth, and provider response

1 **New endpoint pairs carry the rotation and persist.**
Entry contributes $+21.0$ pp; 120-day own-stable shares: USDC 88.3%, USDT 87.7%.

2 **Competitive bridge depth turns support into adoption.**
30-day adoption: 42.6% below $0.1\times$ WETH depth; 84.1% otherwise.

3 **V4 internal routing: incumbents act first, new origins follow.**
First-active origins rise $+0.153^{***}$ in days 31 to 120; persistent volatility adds $+0.318^{***}$.

Dominance forms through market entry, local capital, and provider adjustment.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

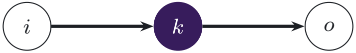
Robustness

- A7 Timing and venue-scope robustness
- A8 Count and value weighting
- A9 Observed route endpoints

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

A1. One reconstructed route is one unit



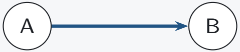
one route unit
two swap legs

- Each linked $i \to k \to o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k.

How we reconstruct a route from one transaction

Direct swap

one transaction

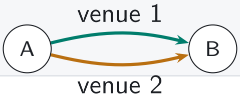


Sequential vehicle route

one transaction

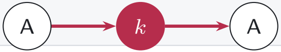
Parallel split

one transaction



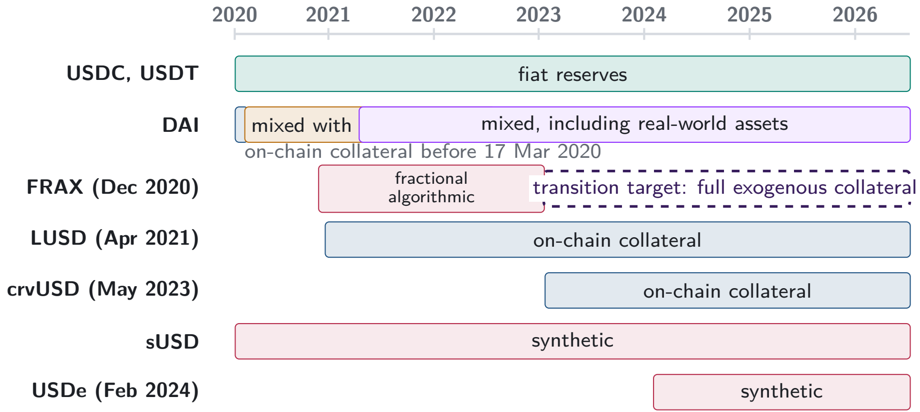
Round trip

one transaction



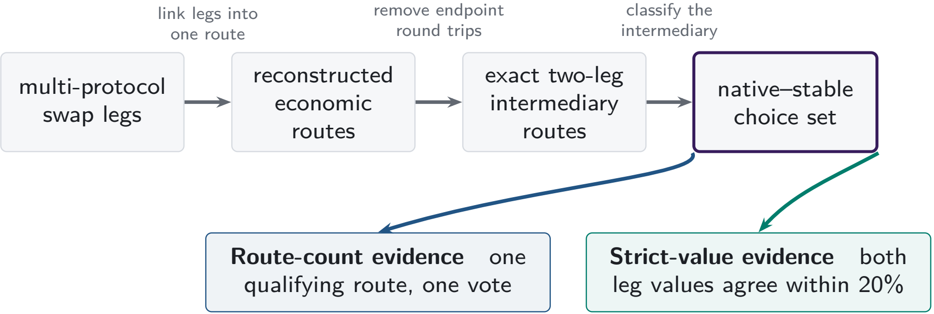
Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level paths make that choice visible inside aggregate turnover.

A2. Stablecoin backing changes over time



Note: Bars begin at public launch or the sample start for older tokens. DAI's dated regimes follow the Maker Governance actions that added USDC-A and activated RWA001-A. FRAX's dashed arrow follows FIP-188 and denotes a target, not observed collateral.

A3. One route universe supports two measurements



Token pairs inside multi-asset pools can supply route legs. V1/V2 architecture evidence is separate; dollar support changes value weighting.

A4. State is reconstructed immediately before execution

1 verified starting state → **2** apply events in blockchain order → **3** recover state before the trade → **4** quote both routes

- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hook-dependent pricing
Constant product	■	–	–	–	–	■	–
Concentrated liquidity	■	■	–	–	–	■	–
StableSwap families	■	–	■	–	■	■	–
Weighted pools	■	–	–	■	■	■	–
V4 singleton	■	■	–	–	–	■	□

Executable-quote validation

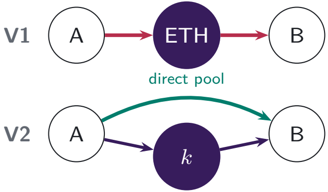
■ recovered pricing input
□ hook-dependent pricing unavailable

Protocol mechanisms studied

V4 singleton flash accounting and netting remain in scope; A5.3 shows the settlement mechanism.

A5.1. Pool formation determines available paths

V1 PROTOCOL RULE $\longrightarrow$ V2 MARKET CHOICE



V1 admits only ETH–token pools. V2 permits any ERC-20 pool, so market creation determines whether direct and vehicle paths coexist.

Which paths exist?

Economic implication: pool creation changes the feasible route set.

A5.2. Executable depth comes from active liquidity

V3 CONCENTRATED-LIQUIDITY POSITIONS

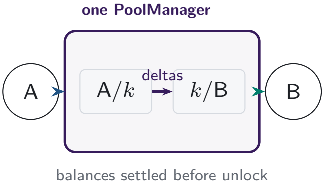
LPs choose a price range and fee tier for each atomic pool. Positions covering the current price determine active liquidity in the direct pool and each vehicle spoke.

Which path offers depth at this trade size?

Economic implication: executable depth is path- and size-specific.

A5.3. Shared accounting moves route settlement

V4 SINGLETON AND LOCK ACCOUNTING



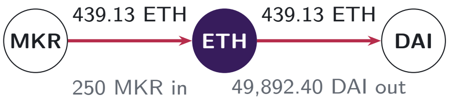
One PoolManager holds all pools. During a lock, the two route legs update balance deltas; the caller settles its obligations before the lock ends.

Where is the route settled?

Economic implication: shared accounting changes the settlement boundary.

A6. V1 forced routes have an on-chain signature

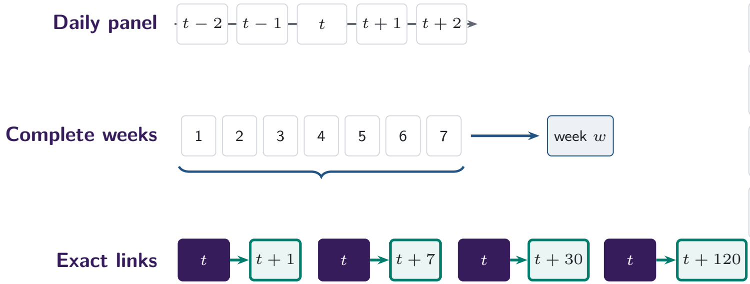
- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.
- 129,810 mandate-era token-to-token transactions carry exactly matching ETH legs; the trace shows the largest.



same transaction, matching ETH flow

Note: Both legs of transaction `0x4dca160d...96a0ca16` (15 March 2020, block 9,674,728) report the same ETH amount to the last recorded digit.

A7. Daily and weekly frequencies answer different questions



Model role

Daily panel date fixed effects

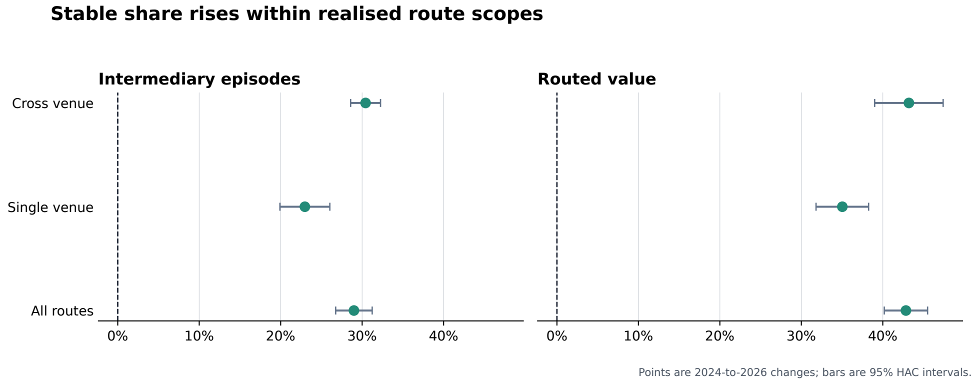
Calendar check week and weekday fixed effects

Complete weeks week fixed effects

Exact links inference by origin date

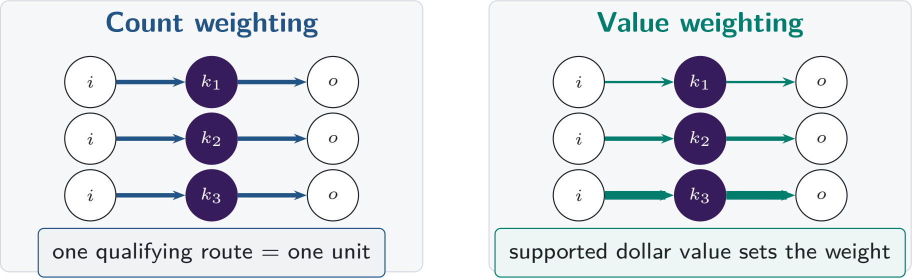
Exact horizons use the required target calendar date.

A7b. Stablecoin use rises within every venue scope



Note: Changes compare matched January-through-June dates in 2024 and 2026. Bars show calendar-HAC confidence intervals; venue scope records realised routes.

A8. Count versus value weighting



Note: Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

Vehicle dominance aggregates realised intermediary choices

One realised route

k occupies the intermediary position

Vehicle dominance

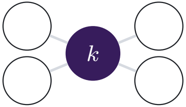
D_k = use through k
$\div$ total intermediary use

Value-weighted routes use comparable dollar amounts

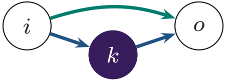
Excess vehicle use

$X_k = D_k \div E_k$
E_k: endpoint share of k

$X_k > 1$: use exceeds the endpoint benchmark



Network position



Execution cost

Endpoint use is the benchmark; network position and same-state cost remain separate.

A9. Intermediary and route-endpoint use are separate

Intermediary share

$$I^N_{k,t} = \frac{n^I_{k,t}}{\sum_j n^I_{j,t}}$$

How often asset k sits inside a route.

Endpoint share

$$E^N_{k,t} = \frac{n^E_{k,t}}{\sum_j n^E_{j,t}}$$

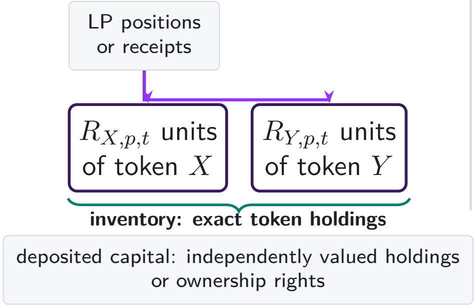
How often asset k appears at either end of the observed pool route.

Observed pool-route endpoints supply the benchmark.

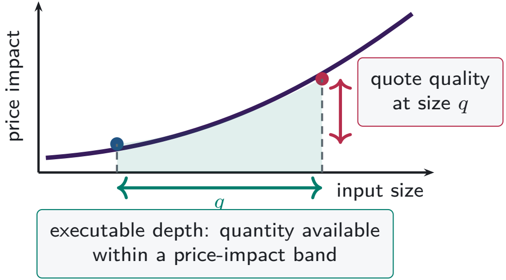
Excess use compares intermediation with observed route-endpoint use on the same support.

A10. Capital, inventory, and depth differ

Pool state at time t

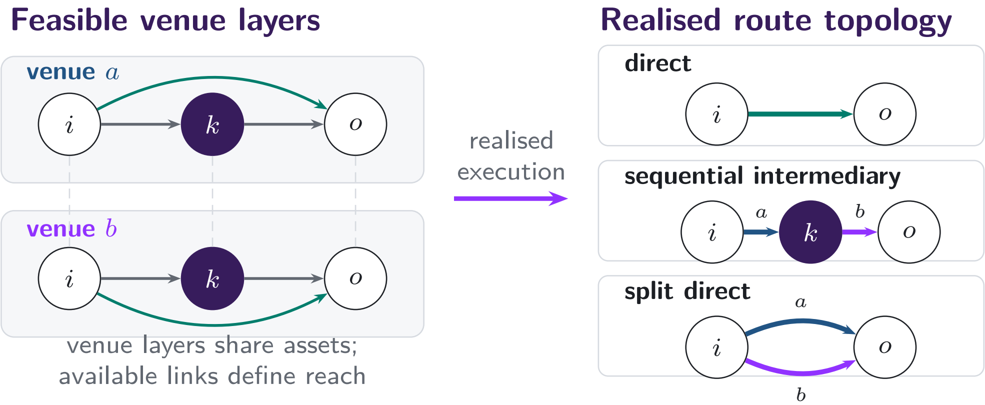


Execution from that state



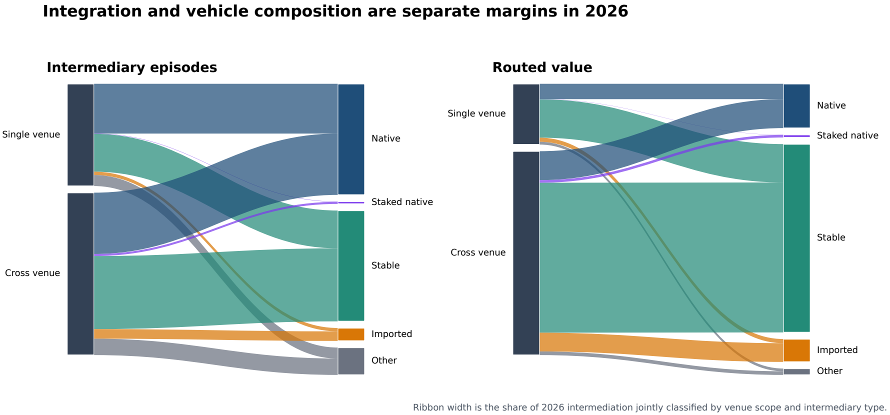
Depth and quotes vary with size and state.

A11. Additional venues expand the feasible route set



venue layers share assets;
available links define reach

A11b. Venue scope and vehicle type differ in 2026



Note: Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

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A13. References: decentralised exchange design

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